

LocateAnything & micro-SAM for EM Segmentation

Three highlights on why they solve different problems — and how they could work together.

July 2026 · Sources: NVIDIA / Hugging Face, arXiv 2605.27365, micro-sam project docs

1 Pixel mask vs. coordinate box — the core mismatch

micro-SAM outputs a **dense pixel mask**; LocateAnything outputs only a **coordinate box** (or point). EM segmentation needs masks, so LocateAnything cannot do it directly. It is also trained purely on natural images, GUIs, OCR and documents — no microscopy — so it cannot even reliably *locate* dense, near-identical EM structures like membranes, mitochondria or vesicles.

The only useful role: **modify the pipeline**. Fine-tune / train LocateAnything on microscopy data → extract bounding boxes → feed those boxes as prompts to **micro-SAM or SAM 3** to produce the accurate pixel-level segmentation. LocateAnything becomes the box detector; the SAM model remains the mask generator.

2 How LocateAnything handles tokenization & bounding boxes

A **Moon-ViT** encoder turns the image into visual tokens at native resolution; a **Qwen2.5** language decoder then emits the location. Instead of streaming free-form numbers, it generates a sequence of fixed-length **box-aligned atomic blocks** of four types:

Semantic — the entity / class

Box — the full (x1,y1,x2,y2) as one unit

Negative — object is absent

End — stop the sequence

Each bounding box is treated as a **single atomic unit of constant length**. Its innovation, **Parallel Box Decoding**, predicts all four coordinates in one parallel step rather than digit-by-digit — keeping the box geometrically coherent and running >10× faster than autoregressive VLMs.

3 Sequential/parallel decoding does not apply to micro-SAM

The "sequential vs. parallel box" question only applies to *token-generative* models. LocateAnything is one; micro-SAM is not.

LOCATEANYTHING

Autoregressive VLM (Qwen2.5 decoder) that emits a token sequence. **Parallel Box Decoding** predicts a whole box in one step instead of token-by-token — an optimization of sequential generation.

MICRO-SAM

Not autoregressive — no token sequence at all. A lightweight mask decoder computes the mask in a **single parallel forward pass**. Nothing to "parallelize" because nothing was sequential.

And the arrow is reversed: a box is LocateAnything's **output** but micro-SAM's **input prompt**.